

PATAAL — Cooling from the Underworld

ACPET Ideathon pitch pack | 20 minutes | Shark Tank format

Working name: PATAAL — *Post-mining Thermal Assets for Affordable Local-cooling* (*pataal* = the deep underworld. Memorable, Indian, and it does the job of the pitch in one word.)

One-line thesis:

India is spending money to fill in its abandoned coal mines. Those voids are the country's largest untapped cooling reservoir — and the people who dug them are the only workforce already trained to run it.

1. The 90-second cold open

(Hold up a lump of coal.)

"For a hundred and fifty years, this is what the mine gave us.

(Put it down. Hold up a block of ice.)

"I want to argue it can give us this instead — and that this is worth more to India in 2047 than that was in 1947.

Here is the problem. India's peak power demand hit 270 GW this May. In 2019 it was around 180 GW. The fastest-growing part of that curve is cooling, and the ugly part is that it now peaks *at night* — after the sun has gone. Night cooling load is the single best argument anyone has for keeping a coal plant open. Cooling is what keeps coal alive.

Here is the asset nobody is looking at. India has identified 147 coal mines for time-bound closure. Forty-two are already scientifically closed. Every one of them is a mapped, surveyed, flooded void, sitting under a district that is heating faster than the national average, with a stranded workforce on top of it. Right now our entire policy imagination for those sites stops at the surface: put solar on it, plant trees, make a park.

Gateshead heats 350 homes from a flooded coal mine. Heerlen in the Netherlands runs a whole town's heating and cooling grid off one, and uses the void as a seasonal battery.

They use the mine for heat. India doesn't need heat. India needs somewhere to *put* heat, and somewhere to *store* cold.

So run it backwards. That's PATAAL."

2. The mechanism (2 minutes — keep it physical, not technical)

Three things a flooded mine void can do that nothing else at that scale can:

1. **Heat sink.** An air-conditioner or chiller in Dhanbad in May is trying to dump heat into 45 °C air. It is terrible at this. Dump it into 28-30 °C mine water instead and the efficiency gain is large — and you stop evaporating water to do it, which matters in a water-stressed district.
2. **Seasonal cold store.** Jharkhand and Chhattisgarh get 8-12 °C winter nights. Charge cold into the void in December, draw it out in May. No battery anyone can build stores energy at that scale for that price.
3. **Night-peak shaving.** Cold stored at noon on solar surplus, used at 11 pm, is load taken directly off the peak that justifies coal. This is a demand-side energy-transition intervention dressed as a cooling business.

Be honest and say it out loud: this is not free cold. It is a heat pump or chiller with a much better sink and a very large store. The claim is COP and storage, not magic. Saying this yourself before a juror says it for you buys enormous credibility.

3. Why this is ACPET's and not IIT's

Map it explicitly onto their verticals. Put this on a slide as a 2x2 — it is the slide that wins the room.

ACPET vertical	What PATAAL contributes
Coal Transition	Extends post-mining land use doctrine <i>below</i> the surface. Natural Trans-Mine Phase III.
People-centric Power Reform	A community-owned cooling utility — the electricity commons argument applied to cold. Plus DISCOM demand-side management value.
Social Impact of Energy Transition	Heat-health outcomes, ex-miner livelihoods, women-operated cold storage, cooling for CHCs and anganwadis. Direct adjacency to SCALE-HI.
Critical Minerals & Circular Economy	Mine water is acid mine drainage. Treating it is a cost you must pay anyway; recovering dissolved metals turns that cost into a co-product.
Energy Futures Lab	Phase 0 is a dashboard. See below.

4. Phase 0: the cheap, fundable first thing (this is what you actually ask for)

Sharks fund a *next step*, not a vision. Ask for a small, concrete, 6-month deliverable.

The India Mine Thermal Atlas. A desk GIS screening layer over every closed and closing coal mine: void geometry and volume from CMPDI abandonment plans, standing water level, depth, water chemistry where available — overlaid with land surface temperature, heat-health burden, and cooling demand within 5 km (hospitals, cold stores, schools, markets).

Outputs:

- Ten ranked candidate sites with a defensible screening methodology.
- One working paper.
- One policy brief to the Ministry of Coal arguing that mine closure guidelines should treat the void as a **thermal asset to be characterised**, not purely a liability to be filled. This is the real policy hook — the AAROH framework currently thinks about the surface only.
- A public EFL dashboard layer.

Ask: roughly ₹25-40 lakh, six months, one hydrogeologist, one GIS analyst, one energy-systems modeller. Phase 1 (instrumented single-site feasibility with a partner mine) follows at roughly ₹1.5-2 crore.

Being able to name a number, a timeline and a team composition is most of what separates a pitch from a daydream.

5. Funding and partnership map (they explicitly asked for this — do not skip it)

Research funding

- **Wellcome Trust** — already an ACPET partner, funds climate-and-health. Heat, cooling access and health outcomes is squarely their window.
- **GEAPP** (Global Energy Alliance for People and Planet) — already a partner; energy access and just transition.
- **HSBC** — founding patron; climate solutions and transition finance.
- **Clean Cooling Collaborative and Sustainable Energy for All (Cooling for All)** — the two obvious cooling-specific funders.
- **NITI Aayog** — ACPET already has a research partnership; route the policy brief through it.

Pilot capital

- **District Mineral Foundation funds** in Dhanbad, Korba, Angul, Ramgarh. Large unspent balances, legally intended for mining-affected communities, and a community cooling utility is an almost perfect fit. Name this specifically; it signals you understand how money actually moves in coal districts.
- **Coal India mine closure escrow funds** and CSR.
- **IREDA / SIDBI** green credit lines; **World Bank / ADB** just-transition windows.

Technical partners

- **CMPDI** — the critical one. They hold the mine abandonment plans and water-level data. Without them there is no atlas.

- **CSIR-CIMFR, Dhanbad** and **IIT (ISM) Dhanbad** — mining engineering and hydrogeology.
- **Central Ground Water Authority** — abstraction clearances.
- **Mining Remediation Authority (UK)** and **Mijnwater BV, Heerlen** — the two organisations on earth with operating experience. A letter of interest from either is cheap and extremely persuasive.
- **BEE** — links to ICAP and efficiency standards.

Commercial model (for the commercial-concept half of the brief) Cooling-as-a-Service: sell cold by the tonne-hour rather than selling equipment, so the hospital or the farmer producer organisation carries no capex. Revenue stacks from cold-storage rental, institutional cooling contracts, DISCOM demand-response payments for verified night-peak reduction, and cooling-efficiency carbon credits. Optional high-margin anchor tenant: a small data centre on reclaimed coalfield land with grid infrastructure already in place and a free heat sink beneath it. Mention this only if a juror pushes on unit economics — it is commercially strong but dilutes the people-centric framing if you lead with it.

6. Slide plan (six slides, no more)

1. **Coal and ice.** Title slide. One image, the thesis sentence.
 2. **The night peak.** 270 GW, the nocturnal shape of the curve, the sentence "cooling is what keeps coal alive."
 3. **The asset.** Map of India: closed/closing mines overlaid on the heat map. Photo of Gateshead or Heerlen next to a photo of a flooded Jharkhand mine.
 4. **Run it backwards.** The three mechanisms — sink, store, peak-shave — as three icons. Plus the honest caveat.
 5. **Who runs it.** Ex-miner skills mapped to operating roles. Community ownership structure. Who gets cooled first.
 6. **The ask.** Phase 0, the number, the timeline, the atlas mock-up, the funder logos.
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7. Props

- **A lump of coal and a block of ice in a cooler.** The whole pitch in two objects. Costs nothing, cannot fail, and is the thing they will remember at lunch.
 - **A printed A3 map** of closed mines over the summer heat map, as a physical mock of the Atlas. Hand it to the jury.
 - Optional and cheap: **a clear jar** layered with black gravel and water, with a loop of clear tubing through it — a cross-section of a flooded void with a heat exchanger in it. Buy at any hardware shop.
 - Skip anything that needs power, pumps or a live demo. Two days is not enough to de-risk a physical demo, and a failed demo costs more than no demo.
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8. Jury Q&A drill — the eight attacks, and your answers

Prepare these as spoken answers, not notes. Two of the four jurors visibly work on critical minerals and on coal/methane, so expect pressure from those directions specifically.

1. **"Mine water is acidic and full of iron. Your heat exchangers will foul in a year."** Correct, and the Gateshead literature flags ochre scaling explicitly. Mitigation is a closed secondary loop and plate exchangers designed for replacement. And the treatment cost is being paid anyway as a closure obligation — we are proposing to attach revenue to it, and potentially metal recovery.
 2. **"Mine water at 30 °C isn't cold. This doesn't work."** Agreed, which is why the claim is a sink and a store, not a source. The comparison is not 30 °C against ice; it is 30 °C against 45 °C ambient air on the condenser side.
 3. **"Why hasn't anyone in India done this?"** Because mine closure policy frames the void as a liability to be filled and the reuse conversation stops at the surface. There is a small Indian geothermal literature on these mines, all framed as power generation, which is the wrong application. Nobody has asked the cooling question.
 4. **"Subsidence. Mine fires. Jharia."** These are screening criteria, not objections. Fire-affected and unstable seams are excluded at Phase 0. The Atlas exists precisely to find the sites where this isn't a problem.
 5. **"Coal towns are low-density. District cooling needs demand density."** True, so we do not start with a district network. We start with a single anchor institution — the district hospital, or a cold store — and prove the physics and the economics at one node.
 6. **"Who pays?"** DMF for capex, cooling-as-a-service tariffs for opex, DISCOM demand-response for the peak value. Named, not hand-waved.
 7. **"Is this research or a company?"** Both, deliberately, in sequence. Phase 0 and 1 are research and produce publishable work plus a policy brief. Phase 2 is a licensable operating model. The research de-risks the company.
 8. **"You are not a mining engineer."** Correct. This is a research programme design and a funding thesis; the hydrogeology is a hire, and CMPDI and CSIR-CIMFR are the partners. ACPET's value-add is not the drilling — it is the people-centric framing, the policy channel and the evidence base.
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9. Two-day schedule

Day 1, morning. Read the four sources in §10 marked ★. Lock the three numbers you will quote: 270 GW peak, 147 mines / 42 closed, eightfold cooling demand growth by 2037-38. Memorise them cold. Do not add a fourth.

Day 1, afternoon. Build the six slides. Write the cold open word for word. Keep everything else as bullets you improvise from.

Day 1, evening. Read the cold open aloud five times. Time it. If it runs over 100 seconds, cut.

Day 2, morning. Buy the props. Print the map. Have someone who knows nothing about energy listen to the pitch and tell you where they got lost.

Day 2, afternoon. Q&A drill. Have someone fire the eight questions in random order. Answer in under 45 seconds each.

Day 2, evening. Two full run-throughs. Target 11-12 minutes of pitch, leaving 8 for the jury. Prepare a 6-minute cut in case they run late — drop slides 4 and 5, keep coal, ice, the map and the ask.

10. References

★ = read these first.

- ★ IEA (June 2026), *India's electricity demand grows at night: Managing rising cooling demand*. <https://www.iea.org/commentaries/india-s-electricity-demand-grows-at-night-managing-rising-cooling-demand>
- ★ Mongabay India (August 2026), *The mine closes but the transition is far from over*. <http://india.mongabay.com/2026/08/the-mine-closes-but-the-transition-is-far-from-over/>
- ★ Ministry of Coal (July 2026), *AAROH — Annual Report on Mine Closure*. Coverage: <https://www.indiablooms.com/finance/big-milestone-india-reclaims-42-abandoned-coal-mines-unveils-aaroh-report/details>
- ★ Banks et al. (2022), *Conceptual Modelling of Two Large-Scale Mine Water Geothermal Energy Schemes: Felling, Gateshead, UK*, IJERPH. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8834720/>
- Verhoeven et al. (2014), *Minewater 2.0 Project in Heerlen*, Energy Procedia. <https://www.sciencedirect.com/science/article/pii/S187661021400174X>
- Menéndez et al. (2019), *Feasibility analysis of using mine water from abandoned coal mines in Spain for heating and cooling of buildings*, Renewable Energy. <https://www.sciencedirect.com/science/article/abs/pii/S0960148119310742>
- Shandong mine water case (2024), *Analysis of geothermal heat recovery from abandoned coal mine water for clean heating and cooling*, Renewable Energy. <https://www.sciencedirect.com/science/article/abs/pii/S0960148124007274>
- Walls et al. (2021) — comprehensive review of mine water geothermal; and Scottish stakeholder study, *Earth Science, Systems and Society*. <https://www.lyellcollection.org/doi/full/10.3389/esss.2024.10121>
- Sircar et al., *Utilization of Abandoned Coal Mines as a Low Enthalpy Geothermal Resource* (SONATA belt, India). https://iqac.pdpu.ac.in/NAAC2/Docs/Criteria%20III/3.4.6/Proofs/CONF/anirbid.sircar_Conf_Utilization%20of%20abandoned%20coal%20mines.pdf
- CLASP (2026), *India Continues Making Major Strides Towards Energy Efficiency — ICAP eightfold cooling demand figure*. <https://www.clasp.ngo/updates/india-making->

major-strides-towards-energy-efficiency/

- SECL mine closure and repurposing programme (August 2026).
<https://www.psuconnect.in/psu-news/secl-completes-scientific-closure-of-28-abandoned-mines>
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11. Backup ideas, if this one doesn't feel like yours

- **Ash to assets.** India's fly ash stock as a critical minerals resource — rare earths and alumina recovery. Sits across Coal Transition and Critical Minerals. Strong, but more chemistry-heavy and closer to existing work.
- **The night peak commons.** Pay coal-district households to pre-cool and thermally store at midday, restructuring the DISCOM tariff as a commons. Cheapest to pilot, least physically exciting.
- **Mine void pumped hydro.** Underground pumped hydro storage in abandoned shafts. Same asset as PATAAL, grid storage instead of cooling. Bigger numbers, much worse people-centric story.

One caution on all of these: avoid coal mine methane. ACPET already runs COMPASS on exactly that, and at least one juror and one co-presenter have published on it.